

PRODUCT INFORMATION

ONDANSETRON-AFT

AFT Pharmaceuticals Ltd.

1. PRODUCT NAME

Ondansetron-AFT Solution for Injection 2mg/mL.
(as Ondansetron hydrochloride dihydrate)

2. PRODUCT DESCRIPTION

Ondansetron-AFT Solution for Injection 2mg/ml is a clear, colorless, transparent solution.

The active ingredient is Ondansetron Hydrochloride dihydrate and available in the 2ml and 4ml ampoule and each ampoule contains the equivalent of 4 mg and 8 mg of Ondansetron respectively.

The solution for injection also contains citric acid, sodium citrate, sodium chloride and water for injection as the excipients.

The reconstituted solution of Ondansetron-AFT solution for injection is also a clear, colorless solution (for reconstitution, see section DOSAGE AND ADMINISTRATION, Instruction for Use/Handling)

3. PHARMACOLOGICAL PROPERTIES

3.1 PHARMACODYNAMIC

Mechanism of Action

Ondansetron is a potent, highly selective 5HT₃ receptor-antagonist. Its precise mode of action in the control of nausea and vomiting is not known.

Chemotherapeutic agents and radiotherapy may cause release of 5HT in the small intestine initiating a vomiting reflex by activating vagal afferents via 5HT₃ receptors. Ondansetron blocks the initiation of this reflex. Activation of vagal afferents may also cause a release of 5HT in the area postrema, located on the floor of the fourth ventricle, and this may also promote emesis through a central mechanism. Thus, the effect of ondansetron in the management of the nausea and vomiting induced by cytotoxic chemotherapy and radiotherapy is probably due to antagonism of 5HT₃ receptors on neurons located both in the peripheral and central nervous system.

The mechanisms of action in post-operative nausea and vomiting are not known but there may be common pathways with cytotoxic induced nausea and vomiting.

Pharmacodynamics effect

Ondansetron does not alter plasma prolactin concentrations.

QT Prolongation

The effect of ondansetron on the QTc interval was evaluated healthy adult men and women. Ondansetron doses included 8 mg and 32 mg infused intravenously over 15 minutes. At the highest tested dose of 32 mg, the maximum mean (upper limit of 90% CI) difference in QTcF from placebo after baseline-correction was 19.6 (21.5) msec. At the lower tested dose of 8 mg, the maximum mean (upper limit of 90% CI) difference in QTcF from placebo after baseline-correction was 5.8 (7.8) msec. There were no QTcF measurements greater than 480 msec and no QTcF prolongation was greater than 60 msec. No significant changes were seen in the measured electrocardiographic PR or QRS intervals.

3.2 PHARMACOKINETIC

Following oral administration, ondansetron is passively and completely absorbed from the gastrointestinal tract and undergoes first pass metabolism. Peak plasma concentrations of about 30 ng/mL are attained approximately 1.5 hours after an 8 mg dose. For doses above 8 mg the increase in ondansetron systemic exposure with dose is greater than proportional; this may reflect some reduction in first pass metabolism at higher oral doses. Mean bioavailability in healthy male subjects, following the oral administration of a single 8 mg tablet, is approximately 55 to 60%. Bioavailability, following oral administration, is slightly enhanced by the presence of food but unaffected by antacids. Studies in healthy elderly volunteers have shown slight, but clinically insignificant, age-related increases in both oral bioavailability (65%) and half-life (5 hours) of ondansetron.

The disposition of ondansetron following oral, intramuscular and intravenous dosing in adults is similar with a terminal half-life of about 3 hours and steady state volume of distribution of about 140 L. Equivalent systemic exposure is achieved after intramuscular and intravenous administration of ondansetron.

A 4mg intravenous infusion of ondansetron given over 5 minutes results in peak plasma concentrations of about 65 ng/mL. Following intramuscular administration of ondansetron, peak plasma concentrations of about 25 ng/mL are attained within 10 minutes of injection.

Following administration of ondansetron suppository, plasma ondansetron concentrations become detectable between 15 and 60 minutes after dosing. Concentrations rise in an essentially linear fashion, until peak concentrations of 20-30 ng/mL are attained, typically 6 hours after dosing. Plasma concentrations then fall, but at a slower rate than observed following oral dosing due to continued absorption of ondansetron. The absolute bioavailability of ondansetron from the suppository is approximately 60% and is not affected by gender. The half-life of the elimination phase following suppository administration is determined by the rate of ondansetron absorption, not systemic clearance and is approximately 6 hours. Females show a small, clinically insignificant, increase in half-life in comparison with males.

Ondansetron is not highly protein bound (70-76%). Ondansetron is cleared from the systemic circulation predominantly by hepatic metabolism through multiple enzymatic pathways. Less than 5% of the absorbed dose is excreted unchanged in the urine. The absence of the enzyme CYP2D6 (the debrisoquine polymorphism) has no effect on ondansetron's pharmacokinetics. The pharmacokinetic properties of ondansetron are unchanged on repeat dosing.

Special Patient Populations

Gender

Gender differences were shown in the disposition of ondansetron, with females having a greater rate and extent of absorption following an oral dose and reduced systemic clearance and volume of distribution (adjusted for weight).

Children and Adolescents (aged 1 month to 17 years)

In paediatric patients aged 1 to 4 months (n=19) undergoing surgery, weight normalised clearance was approximately 30% slower than in patients aged 5 to 24 months (n=22) but comparable to the patients aged 3 to 12 years. The half-life in the patient population aged 1 to 4 month was reported to average 6.7 hours compared to 2.9 hours for patients in the 5 to 24 month and 3 to 12 year age range. The differences in pharmacokinetic parameters in the 1 to 4 month patient population can be explained in part by the higher percentage of total body water in neonates and infants and a higher volume of distribution for water soluble drugs like ondansetron.

In paediatric patients aged 3 to 12 years undergoing elective surgery with general anaesthesia, the absolute values for both the clearance and volume of distribution of ondansetron were reduced in comparison to values with adult patients. Both parameters increased in a linear fashion with weight and by 12 years of age, the values were approaching those of young adults. When clearance and volume of distribution values were normalised by body weight, the values for these parameters were similar between the different age group populations. Use of weight-based dosing compensates for age-related changes and is effective in normalising systemic exposure in paediatric patients.

Population pharmacokinetic analysis was performed on 428 subjects (cancer patients, surgery patients and healthy volunteers) aged 1 month to 44 years following intravenous administration of ondansetron. Based on this analysis, systemic exposure (AUC) of ondansetron following oral or IV dosing in children and adolescents was comparable to adults, with the exception of infants aged 1 to 4 months. Volume was related to age and was lower in adults than in infants and children. Clearance was related to weight but not to age with the exception of infants aged 1 to 4 months. It is difficult to conclude whether there was an additional reduction in clearance related to age in infants 1 to 4 months or simply inherent variability due to the low number of subjects studied in this age group. Since patients less than 6 months of age will only receive a single dose in PONV a decreased clearance is not likely to be clinically relevant.

Elderly

Early Phase I studies in healthy elderly volunteers showed a slight age-related decrease in clearance, and an increase in half-life of ondansetron. However, wide inter-subject variability resulted in considerable overlap in pharmacokinetic parameters between young (< 65 years of age) and elderly subjects (≥ 65 years of age) and there were no overall differences in safety or efficacy observed between young and elderly cancer patients enrolled in CINV clinical trials to support a different dosing recommendation for the elderly.

Based on more recent ondansetron plasma concentrations and exposure-response modelling, a greater effect on QTcF is predicted in patients ≥ 75 years of age compared to young adults.

Specific dosing information is provided for patients over 65 years of age and over 75 years of age (see section DOSAGE AND ADMINISTRATION).

Renal Impairment

In patients with renal impairment (creatinine clearance 15-60 mL/min), both systemic clearance and volume of distribution are reduced following intravenous administration of ondansetron, resulting in a slight, but clinically insignificant, increase in elimination half-life (5.4 hours). A study in patients with severe renal impairment who required regular haemodialysis (studied between dialyses) showed ondansetron's pharmacokinetics to be essentially unchanged following intravenous administration.

Hepatic Impairment

Following oral, intravenous or intramuscular dosing in patients with severe hepatic impairment, ondansetron's systemic clearance is markedly reduced with prolonged elimination half-lives (15 to 32 hours) and an oral bioavailability approaching 100% due to reduced pre-systemic metabolism. The pharmacokinetics of ondansetron following administration as a suppository have not been evaluated in patients with hepatic impairment.

3.3 Pre-clinical Safety Data

No additional data of relevance.

4. INDICATIONS

Adults:

Ondansetron-AFT injection is indicated for the management of nausea and vomiting induced by cytotoxic chemotherapy and radiotherapy. Ondansetron-AFT injection is also indicated for the prevention and treatment of post-operative nausea and vomiting (PONV)

Paediatric Population:

Ondansetron-AFT is indicated for the management of nausea and vomiting induced by cytotoxic chemotherapy (CINV)

5. DOSAGE AND ADMINISTRATION

Chemotherapy and Radiotherapy:

Adults:

The emetogenic potential of cancer treatment varies according to the doses and combinations of chemotherapy and radiotherapy regimens used. The route of administration and dose of Ondansetron-AFT should be flexible in the range of 8-32mg a day and selected as shown below.

Emetogenic chemotherapy and radiotherapy: Ondansetron-AFT can be given either by rectal, oral (tablets or syrup), intravenous or intramuscular administration.

For most patients receiving emetogenic chemotherapy or radiotherapy, Ondansetron-AFT 8mg should be administered as a slow intravenous injection (in not less than 30 seconds) or intramuscular injection, immediately before treatment, followed by 8mg orally twelve hourly.

To protect against delayed or prolonged emesis after the first 24 hours, oral or rectal treatment with Ondansetron-AFT should be continued for up to 5 days after a course of treatment.

Highly emetogenic chemotherapy: For patients receiving highly emetogenic chemotherapy, e.g high-dose cisplatin, Ondansetron-AFT can be given either by oral, rectal, intravenous or intramuscular administration. Ondansetron-AFT has been shown to be equally effective in the following dose schedules over the first 24 hours of chemotherapy:

- A single dose of 8mg by slow intravenous injection (in not less than 30 seconds) or intramuscular injection immediately before chemotherapy.
- A dose of 8mg by slow intravenous injection (in not less than 30 seconds) or intramuscular injection immediately before chemotherapy, followed by two further intravenous injection(in not less than 30 seconds) or intramuscular doses of 8mg four hours apart, or by a constant infusion of 1mg/hour for up to 24 hours.
- A maximum initial intravenous dose of 16mg diluted in 50-100mL of saline or other compatible infusion fluid (see section **Instruction for Use/Handling**) and infused over not less than 15 minutes immediately before chemotherapy. The initial dose of Ondansetron-AFT may be followed by two additional 8mg intravenous doses (in not less than 30 seconds) or intramuscular doses four hours apart.

A single dose greater than 16 mg must not be given due to dose dependent increase of QT-prolongation risk (see sections WARNINGS AND PRECAUTIONS, SIDE EFFECTS AND ADVERSE REACTIONS and PHARMACODYNAMICS)

The selection of dose regimen should be determined by the severity of the emetogenic challenge.

The efficacy of Ondansetron-AFT in highly emetogenic chemotherapy may be enhanced by the addition of a single intravenous dose of dexamethasone sodium phosphate, 20mg administered prior to chemotherapy.

To protect against delayed or prolonged emesis after the first 24 hours, oral or rectal treatment with Ondansetron-AFT should be continued for up to 5 days after a course of treatment.

Paediatric Population:

CINV in children aged ≥ 6 months and adolescents

The dose for CINV can be calculated based on body surface area (BSA) or weight – see below. In paediatric clinical studies, ondansetron was given by IV infusion diluted in 25 to 50mL of saline or other compatible infusion fluid and infused over not less than 15 minutes. Weight-based dosing results in higher total daily doses compared to BSA-based dosing (see sections WARNINGS AND PRECAUTIONS and PHARMACODYNAMICS).

Ondansetron-AFT injection should be diluted in 5% dextrose or 0.9% sodium chloride or other compatible infusion fluid (see section **Instruction for Use/Handling** below) and infused intravenously over not less than 15 minutes.

There are no data from controlled clinical trials on the use of ondansetron in the prevention of delayed or prolonged CINV. There are no data from controlled clinical trials on the use of ondansetron for radiotherapy-induced nausea and vomiting in children.

Dosing by BSA:

Ondansetron-AFT should be administered immediately before chemotherapy as a single intravenous dose of 5 mg/m². The single intravenous dose must not exceed 8 mg.

Oral dosing can commence twelve hours later and may be continued for up to 5 days (Table 2). The total dose over 24 hours (given as divided doses) must not exceed adult dose of 32 mg.

Table 2: BSA-based dosing for Chemotherapy - Children aged 6 months and adolescents

BSA	Day 1 (a,b)	Days 2-6 ^(b)
< 0.6 m ²	5 mg/m ² i.v. plus 2 mg syrup after 12 hours	2 mg syrup every 12 hours
≥ 0.6 m ²	5 mg/m ² i.v. plus 4 mg syrup or tablet after 12	4 mg syrup or tablet every 12 hours
> 1.2 m ²	5mg/m ² or 8mg i.v plus 8mg syrup or tablet after 12 hours	8mg syrup or tablet every 12 hours

a The intravenous dose must not exceed 8mg.

b The total dose over 24 hours must not exceed adult dose of 32 mg

Dosing by body weight:

Weight-based dosing results in higher total daily doses compared to BSA-based dosing (see sections WARNINGS AND PRECAUTIONS and PHARMACODYNAMICS).

Ondansetron-AFT should be administered immediately before chemotherapy as a single intravenous dose of 0.15 mg/kg.

The single intravenous dose must not exceed 8 mg.

Two further intravenous doses may be given in 4-hourly intervals.

Oral dosing can commence twelve hours later and may be continued for up to 5 days (Table 3). The total dose over 24 hours (given as divided doses) must not exceed adult dose of 32 mg.

Table 3: Weight-based dosing for Chemotherapy - Children aged ≥ 6 months and adolescents

Weight	Day 1 (a,b)	Days 2-6 ^(b)
≤ 10 kg	Up to 3 doses of 0.15 mg/kg every 4 hours	2 mg syrup every 12 hours
> 10 kg	Up to 3 doses of 0.15 mg/kg every 4 hours	4 mg syrup or tablet every 12 hours

a The intravenous dose must not exceed 8mg.

b The total dose over 24 hours must not exceed adult dose of 32 mg.

Elderly:

Elderly patients aged 75 years or older: A single dose of intravenous ondansetron given for the prevention of chemotherapy-induced nausea and vomiting (CINV) must not exceed 8mg (infused over at least 15 minutes).

Adult patients aged less than 75 years: A single dose of intravenous ondansetron given for the prevention of CINV in adults (aged less than 75 years) must not exceed 16mg (infused over at least 15 minutes).

Repeat dosing in all adult patients (including elderly patients):

Repeat intravenous doses of ondansetron for the prevention of CINV should be given no less than 4 hours apart.

Dilution and administration in elderly patients aged 65 years or older: All intravenous doses of ondansetron for the prevention of CINV should be diluted in 50–100mL saline or other compatible fluid (see **Instruction for Use/Handling** below) and infused over at least 15 minutes.

Patients with Renal Impairment:

No alteration of daily dosage or frequency of dosing, or route of administration are required.

Patients with Hepatic Impairment:

Clearance of Ondansetron-AFT is significantly reduced and serum half-life significantly prolonged in subjects with moderate or severe impairment of hepatic function. In such patients a total daily dose of 8mg should not be exceeded and therefore parenteral or oral administration is recommended.

Patients with Poor Sparteine/Debrisoquine Metabolism:

The elimination half-life of ondansetron is not altered in subjects classified as poor metabolisers of sparteine and debrisoquine. Consequently in such patients repeat dosing will give drug exposure levels no different from those of the general population. No alteration of daily dosage or frequency of dosing is required.

Post-Operative Nausea and Vomiting (PONV):***Adults:***

For the prevention of PONV Ondansetron-AFT can be administered orally or by intravenous or intramuscular injection.

Ondansetron-AFT may be administered as a single dose of 4mg given by intramuscular or slow intravenous injection at induction of anaesthesia.

For treatment of established PONV a single dose of 4mg given by intramuscular or slow intravenous injection is recommended.

Paediatric population:**PONV in children aged ≥ 1 month and adolescents**

For prevention of PONV in paediatric patients having surgery performed under general anaesthesia, a single dose of Ondansetron-AFT may be administered by slow intravenous injection (not less than 30 seconds) at a dose of 0.1mg/kg up to a maximum of 4mg either prior to, at or after induction of anaesthesia.

For the treatment of PONV after surgery in paediatric patients having surgery performed under general anaesthesia, a single dose of Ondansetron-AFT may be administered by slow intravenous injection (not less than 30 seconds) at a dose of 0.1mg/kg up to a maximum of 4mg.

There are no data on the use of ondansetron in the treatment of PONV in children below 2 years of age.

Elderly:

There is limited experience in the use of ondansetron in the prevention and treatment of PONV in the elderly, however ondansetron is well tolerated in patients over 65 years receiving chemotherapy.

Patients with Renal Impairment:

No alteration of daily dosage or frequency of dosing, or route of administration are required.

Patients with Hepatic Impairment:

Clearance of ondansetron is significantly reduced and serum half-life significantly prolonged in subjects with moderate or severe impairment of hepatic function. In such patients a total daily dose of 8mg should not be exceeded and therefore parenteral or oral administration is recommended.

Patients with poor Sparteine/Debrisoquine Metabolism:

The elimination half-life of ondansetron is not altered in subjects classified as poor metabolisers of sparteine and debrisoquine. Consequently in such patients repeat dosing will give drug exposure levels no different from those of the general population. No alteration of daily dosage or frequency of dosing are required.

Instruction for Use/Handling

Ondansetron-AFT Injection should not be autoclaved. Compatibility with intravenous fluids Ondansetron-AFT injection should only be mixed with those infusion solutions which are recommended:

- 0.9%w/v Sodium Chloride Intravenous Infusion BP
- 5%w/v Glucose Intravenous Infusion BP
- 10%w/v Mannitol Intravenous Infusion BP
- Ringers Intravenous Infusion
- 0.3%w/v Potassium Chloride and 0.9%w/v Sodium Chloride Intravenous Infusion BP
- 0.3%w/v Potassium Chloride and 5%w/v Glucose Intravenous Infusion BP

In keeping with good pharmaceutical practice dilutions of Ondansetron-AFT injection in intravenous fluids should be prepared at the time of infusion or stored at 2-8 °C for no more than 24 hours before the start of administration.

Compatibility studies have been undertaken in polyvinyl chloride infusion bags and polyvinyl chloride administration sets. It is considered that adequate stability would also be conferred by the use of polyethylene infusion bags or Type 1 glass bottles. Dilutions of ondansetron injection in sodium chloride 0.9%w/v or in glucose 5%w/v have been demonstrated to be stable in polypropylene syringes. It is considered that ondansetron injection diluted with other compatible infusion fluids would be stable in polypropylene syringes.

Compatibility with other drugs: Ondansetron-AFT may be administered by intravenous infusion at 1mg/hour, e.g. from an infusion bag or syringe pump. The following drugs may be administered via the Y-site of the ondansetron injection giving set for ondansetron concentrations of 16 to 160 micrograms/mL (e.g. 8 mg/500 mL and 8 mg/50 mL respectively);

Cisplatin: Concentrations up to 0.48 mg/mL (e.g. 240 mg in 500 mL) administered over one to eight hours.

5-Fluorouracil: Concentrations up to 0.8 mg/mL (e.g. 2.4 g in 3 litres or 400 mg in 500 mL) administered at a rate of at least 20 mL per hour (500 mL per 24 hours). Higher concentrations of 5-fluorouracil may cause precipitation of ondansetron. The 5-fluorouracil infusion may contain up to 0.045%w/v magnesium chloride in addition to other excipients shown to be compatible.

Carboplatin: Concentrations in the range 0.18 mg/mL to 9.9 mg/mL (e.g. 90 mg in 500 mL to 990 mg in 100 mL), administered over ten minutes to one hour.

Etoposide: Concentrations in the range 0.14 mg/mL to 0.25 mg/mL (e.g. 72 mg in 500 mL to 250 mg in 1 litre), administered over thirty minutes to one hour.

Ceftazidime: Doses in the range 250 mg to 2000 mg reconstituted with Water for Injections BP as recommended by the manufacturer (e.g. 2.5 mL for 250 mg and 10 mL for 2g ceftazidime) and given as an intravenous bolus injection over approximately five minutes.

Cyclophosphamide: Doses in the range 100 mg to 1g, reconstituted with Water for Injections BP, 5 mL per 100 mg cyclophosphamide, as recommended by the manufacturer and given as an intravenous bolus injection over approximately five minutes.

Doxorubicin: Doses in the range 10-100 mg reconstituted with Water for Injections BP, 5 mL per 10 mg doxorubicin, as recommended by the manufacturer and given as an intravenous bolus injection over approximately 5 minutes.

Dexamethasone: Dexamethasone sodium phosphate 20mg may be administered as a slow intravenous injection over 2-5 minutes via the Y-site of an infusion set delivering 8 or 16mg of ondansetron diluted in 50-100 mL of a compatible infusion fluid over approximately 15 minutes. Compatibility between dexamethasone sodium phosphate and ondansetron has been demonstrated supporting administration of these drugs through the same giving set resulting in concentrations in line of 32 microgram - 2.5mg/ mL for dexamethasone sodium phosphate and 8 microgram - 1mg/mL for ondansetron.

6. CONTRAINDICATIONS

Concomitant use with apomorphine (see section DRUG INTERACTIONS).

Hypersensitivity to any component of the preparation.

7. WARNINGS AND PRECAUTIONS

Hypersensitivity reactions have been reported in patients who have exhibited hypersensitivity to other selective 5HT₃ receptor antagonists.

Respiratory events should be treated symptomatically and clinicians should pay particular attention to them as precursors of hypersensitivity reactions.

Ondansetron prolongs the QT interval in a dose-dependent manner (see section PHARMACODYNAMICS). In addition, post-marketing cases of Torsade de Pointes have been reported in patients using ondansetron. Avoid ondansetron in patients with congenital long QT syndrome. Ondansetron Should be administered with caution to patients who have or may develop prolongation of QTc, including patients with electrolyte abnormalities, congestive heart failure, bradyarrhythmias or patients taking other medicinal products that lead to QT prolongation or electrolyte abnormalities.

Hypokalaemia and hypomagnesaemia should be corrected prior to ondansetron administration.

There have been post-marketing reports describing patients with serotonin syndrome (including altered mental status, autonomic instability and neuromuscular abnormalities) following the concomitant use of ondansetron and other serotonergic drugs (including selective serotonin reuptake inhibitors (SSRI) and serotonin noradrenaline reuptake inhibitors (SNRIs)). If concomitant treatment with ondansetron and other serotonergic drugs is clinically warranted, appropriate observation of the patient is advised.

As ondansetron is known to increase large bowel transit time, patients with signs of sub-acute intestinal obstruction should be monitored following administration.

In patients with adenotonsillar surgery prevention of nausea and vomiting with ondansetron may mask occult bleeding. Therefore, such patients should be followed carefully after ondansetron.

Paediatric Population:

Paediatric patients receiving ondansetron with hepatotoxic chemotherapeutic agents should be monitored closely for impaired hepatic function.

CINV

When calculating the dose on an mg/kg basis and administering three doses at 4-hour intervals, the total daily dose will be higher than if one single dose of 5mg/m² followed by an oral dose is given. The comparative efficacy of these two different dosing regimens has not been investigated in clinical trials. Cross-trial comparison indicates similar efficacy for both regimens (section PHARMACODYNAMICS).

Effects on Ability to Drive and Use Machines

In psychomotor testing ondansetron does not impair performance nor cause sedation. No detrimental effects on such activities are predicted from the pharmacology of ondansetron.

8. USE IN PREGNANCY AND LACTATION

Pregnancy

The safety of ondansetron for use in human pregnancy has not been established. Evaluation of experimental animal studies does not indicate direct or indirect harmful effects with respect to the development of the embryo, or foetus, the course of gestation and peri- and post-natal development. However as animal studies are not always predictive of human response the use of ondansetron in pregnancy is not recommended.

Breast-feeding

Tests have shown that ondansetron passes into the milk of lactating animals. It is therefore recommended that mothers receiving ondansetron should not breast-feed their babies.

9. DRUG INTERACTIONS

There is no evidence that ondansetron either induces or inhibits the metabolism of other drugs commonly co-administered with it. Specific studies have shown that there are no interactions when ondansetron is administered with alcohol, temazepam, furosemide, alfentanil, tramadol, morphine, lidocaine, thiopental, or propofol.

Ondansetron is metabolised by multiple hepatic cytochrome P-450 enzymes: CYP3A4, CYP2D6 and CYP1A2.

Due to the multiplicity of metabolic enzymes capable of metabolising ondansetron, enzyme inhibition or reduced activity of one enzyme (e.g. CYP2D6 genetic deficiency) is normally compensated by other enzymes and should result in little or no significant change in overall ondansetron clearance or dose requirement.

Caution should be exercised when ondansetron is coadministered with drugs that prolong the QT interval and/or cause electrolyte abnormalities. (see section WARNINGS AND PRECAUTIONS)

Use of ondansetron with QT prolonging drugs may result in additional QT prolongation. Concomitant use of ondansetron with cardiotoxic drugs (e.g. anthracyclines (such as doxorubicin, daunorubicin) or trastuzumab), antibiotics (such as erythromycin), antifungals (such as ketoconazole), antiarrhythmics (such as amiodarone) and beta blockers (such as atenolol or timolol) may increase the risk of arrhythmias. (See section WARNINGS AND PRECAUTIONS).

Serotonergic Drugs (e.g. SSRIs and SNRIs): There have been post-marketing reports describing patients with serotonin syndrome (including altered mental status, autonomic instability and neuromuscular abnormalities) following the concomitant use of ondansetron and other serotonergic drugs (including SSRIs and SNRIs). (See section WARNINGS AND PRECAUTIONS)

Apomorphine: Based on reports of profound hypotension and loss of consciousness when ondansetron was administered with apomorphine hydrochloride, concomitant use with apomorphine is contraindicated.

Phenytoin, Carbamazepine and Rifampicin: In patients treated with potent inducers of CYP3A4 (i.e. phenytoin, carbamazepine, and rifampicin), the oral clearance of ondansetron was increased and ondansetron blood concentrations were decreased.

Tramadol: Data from small studies indicate that ondansetron may reduce the analgesic effect of tramadol.

10. SIDE EFFECTS AND ADVERSE REACTIONS

Adverse events are listed below by system organ class and frequency.

The following frequencies are estimated at the standard recommended doses of ondansetron. The adverse event profiles in children and adolescents were comparable to that seen in adults.

Immune system disorders	Rare: Immediate hypersensitivity reactions sometimes severe, including anaphylaxis.
Nervous system disorders	Very common: Headache. Uncommon: Seizures, movement disorders (including extrapyramidal reactions such as dystonic reactions, oculogyric crisis and dyskinesia) ⁽¹⁾ . Rare: Dizziness during rapid IV administration.
Eye disorders	Rare: Transient visual disturbances (e.g. blurred vision) predominantly during IV administration. Very rare: Transient blindness predominantly during intravenous administration. ⁽²⁾
Cardiac disorders	Uncommon: Arrhythmias, chest pain with or without ST segment depression, bradycardia. Rare: QTc prolongation (including Torsade de Pointes)
Vascular disorders	Common: Sensation of warmth or flushing. Uncommon: Hypotension.
Respiratory, thoracic and mediastinal disorders	Uncommon: Hiccups.
Gastrointestinal disorders	Common: Constipation.
Hepatobiliary disorders	Uncommon: Asymptomatic increases in liver function tests. ⁽³⁾
General disorders and administration site conditions	Common: Local IV injection site reactions.

(1) Observed without definitive evidence of persistent clinical sequelae.

(2) The majority of the blindness cases reported resolved within 20 minutes. Most patients had received chemotherapeutic agents, which included cisplatin. Some cases of transient blindness were reported as cortical in origin.

(3) These events were observed commonly in patients receiving chemotherapy with cisplatin.

11. OVERDOSE AND TREATMENTS

Symptoms and Signs

There is limited experience of ondansetron overdose. In the majority of cases, symptoms were similar to those already reported in patients receiving recommended doses (see section SIDE EFFECTS AND ADVERSE REACTIONS). Manifestations that have been reported include visual disturbances, severe constipation, hypotension and a vasovagal episode with transient second-degree AV block.

Ondansetron prolongs the QT interval in a dose-dependent fashion. ECG monitoring is recommended in cases of overdose.

Treatment

There is no specific antidote for ondansetron, therefore in all cases of suspected overdose, symptomatic and supportive therapy should be given as appropriate.

The use of ipecacuanha to treat overdose with ondansetron is not recommended, as patients are unlikely to respond due to the anti-emetic action of ondansetron itself.

12. INCOMPATIBILITIES

Ondansetron injection should not be administered in the same syringe or infusion as any other medication. Ondansetron injection should only be mixed with those infusion solutions that are recommended.

13. SHELF LIFE AND STORAGE CONDITIONS

Shelf life:

2 years (24 months) from the date of manufacture (unopened).

Storage:

Store below 30 °C. Protect from light.

If not used immediately, keep reconstituted dilution at 2-8 °C for not more than 24 hours.

Dilutions of Ondansetron-AFT injection in compatible intravenous infusion fluids are stable under normal room lighting conditions or daylight for at least 24 hours, thus no protection from light is necessary while infusion takes place.

14. DOSAGE FORMS AND PACKAGING AVAILABLE

Ondansetron-AFT is filled in a clear, colorless Type I glass snap-ring ampoules.

It is available in a 2ml (4mg) and 4ml (8mg) pack size.

5 or 10 ampoules are packed in a carton.

(NOT ALL PRESENTATION ARE AVAILABLE LOCALLY)

15. MANUFACTURER

Qilu Pharmaceuticals Co.,Ltd.

No. 317, Xinluo Road, High-Tech Zone, Jinan, Shandong Province, 250101, China

16. PRODUCT OWNER

In New Zealand:

AFT Pharmaceuticals Ltd.

Level 1, Nielsen Building, 129 Hurstmere Road, Takapuna, Auckland 0622, New Zealand

In Australia:

AFT Pharmaceuticals Pty Ltd,

113 Wicks Road, North Ryde, Sydney, NSW 2113, Australia

17. PRODUCT LICENSE/REGISTRATION HOLDER

For Singapore:

AFT Pharmaceuticals Singapore Pte. Ltd.

16 Raffles Quay, #33-03 Hong Leong Building, Singapore 048581

For Malaysia:

AFT Pharmaceuticals (SE Asia) Sdn. Bhd.

A-29-3, Level 29, Menara UOA Bangsar, No. 5 Jalan Bangsar Utama 1, 59000 Kuala Lumpur, Malaysia

18. REGISTRATION NUMBERS

Singapore: SIN15006P

Malaysia: MALxxxxxxxACZ

19. DATE OF REVISION: November 2016